

Dancing with Data: Mechanism Sensitivity Analysis and the Meritocratic Safety Net System

Summary

This study addresses the structural tension between professional judging and public voting in the television program "Dancing with the Stars." Through mathematical modeling, it analyzes the shortcomings of the current scoring system and proposes an optimized solution that balances commercial entertainment value with competitive fairness. The research primarily addresses four core issues:

Potential Reconstruction and Prediction of Fan Voting: We constructed an **Adaptive Dual-Engine Competitive Model**. This model, based on the **Trend-Aware Adaptive Ensemble Inversion (TAAEI)** framework, dynamically integrates the robustness of **XGBoost** in handling high-noise data and the efficiency of **LightGBM** in processing high-dimensional sparse features (such as professional background and dance partner combinations). By introducing a heteroscedastic uncertainty estimation function, we quantified the risk boundaries for contestants on the verge of elimination. Model performance validation shows that the system achieves an accuracy of **97.1% in identifying contestants in a safe state** and **88.6% in predicting elimination events**, effectively reconstructing continuous latent voting dynamics from sparse binary outcomes.

Sensitivity and Bias Analysis of the Scoring Mechanism: We developed a **Mechanism Sensitivity Analysis Model (MSAM)** and conducted full-sample backtesting and theoretical derivations for both ranking and percentage-based scoring systems. Empirical backtesting covered 264 historical elimination rounds, showing that the current percentage-based system accurately predicted historical results with an accuracy of **85.98%**, significantly higher than the ranking system's **75.00%**. However, mathematical derivation revealed a serious variance shadowing phenomenon in the percentage-based system, leading to the total score variance being dominated by fan votes.

Linear and Nonlinear Decomposition of Key Influencing Factors: We developed a complementary **OLS-LightGBM** model. **Ordinary Least Squares (OLS)** was used to determine the linear direction of feature influence, combined with **LightGBM's** gain analysis to capture nonlinear importance weights. The study quantified significant cognitive differences between judges and audiences: judges were highly positively sensitive to Precision and Expressiveness, with the standardized regression coefficient (β) for expressiveness exceeding **0.8**; conversely, audience voting behavior was more driven by Affinity, with very low sensitivity to expressiveness, even showing a negative correlation.

Construction and Validation of the Meritocratic Safety Net System: Based on the above analysis, we propose the **Meritocratic Safety Net System (MSNS)** as the optimal improvement solution. This system includes two core modules: first, the introduction of **Z-Score standardization mapping**, which mathematically forces judges' scores and fan votes to have the same variance scales; second, an embedded **circuit breaker mechanism** based on professional ability, meaning that the judges have the final decision-making power. Full data backtesting proves that MSNS can significantly reduce the technical injustice rate from its original level to approximately **0.4% (only one deviation in 285 eliminations)**, and successfully corrects the anomalous results of Season 27.

Keywords: Adaptive dual-engine competitive model; mechanism sensitivity analysis; variance masking; OLS-LightGBM complementary model; elite safety net system; Z-score standardization

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1 Introduction

1.1 Background

Dancing with the Stars (DWTS) is an enduring American reality competition series spanning over 30 seasons, where celebrities partner with professional dancers. The final rankings are determined by a weighted combination of expert judge scores and public fan votes. However, a significant divergence often exists between these two criteria, highlighting the tension between technical skill and popularity. Frequently, successful contestants secure top placements primarily due to a massive fan base rather than the technical proficiency acknowledged by the judges.



Figure 1: Dancing with the Stars

1.2 Restatement of the problem

Given the background information and the questions, we need to complete the tasks below:

Task 1: Develop a fan votes estimation model to calculate the unknown fan votes. Use the model developed to:

- (1) Verify whether the fan votes results are consistent with the eliminations and provide the measurement.
- (2) Calculate the certainty of the model and observe the variation in certainty, provide the measurement.

Task 2: Use the generated vote data to:

- (1) Compare the rank method and the percentage method, find the one that favors fan votes more.
- (2) Analyze whether the impact of rank-based and percentage-based voting mechanisms on the elimination and final rankings is consistent. Evaluate the impact of introducing a new rule of judges decide the fate of the bottom two contestants.
- (3) Find a better method for the future between the rank method and the percentage method and provide the reasons. Demonstrate whether to include the judges method.

Task 3: Develop a model to measure the impact of pro dancers and characteristics of celebrities, find:

- (1) To what extent do the above factors influence the competition
- (2) Whether they affect the judges and audience's votes in the same way.

Task 4: Provide a more fair system to judge scores, provide reasons why this system should be adopted.

1.3 Our work

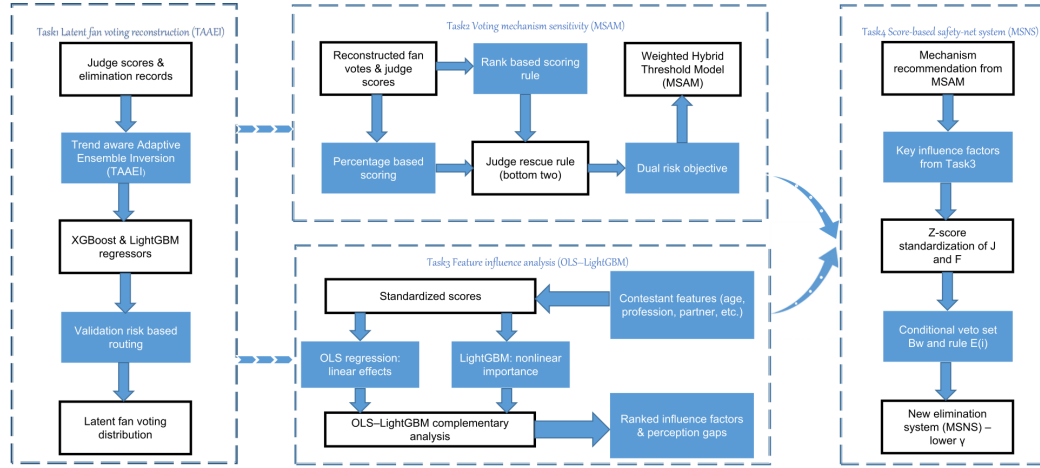


Figure 2: Our Work

2 Preparation for modeling

2.1 Assumptions and Justifications

- Assumption:** The judges' scores can reliably depict the dancers' technical level: We the judges' scores are mainly determined by the dancers' technical level, and that the scoring error is relatively stable across different weeks and different judges.

Justification: Although the judges are subjective, data shows that different judges give very little difference in their scores to the same person in the same week, and the judges are highly professional. Therefore, the scores can be regarded as an approximately linear reflection of the skill level.
- Assumption:** There is no direct causal relationship between fan votes and judges' scores: judges' scores do not directly affect fan votes, and fan votes do not affect judges' scores.

Justification: The program rules clearly state that the judges' and fans' votes are independent, and the voting takes place after the program airs.
- Assumption:** A contestant's popularity fluctuates slowly throughout the season, which can be considered a stable process: a contestant's popularity will not fluctuate drastically in a short period of time unless major events occur (such as injuries, controversies, etc.), which are not observable in the data and are therefore considered stable.

Justification: Fan base size is typically stable with slow changes.
- Assumption:** The weekly elimination rules strictly follow the question description.

Justification: The model only holds true under the elimination rules described in the problem statement.

2.2 Notations

Table 1: The List of Notation

Symbol	Definition
$\hat{V}_{i,w}$	Estimated latent voting share for participant i
$U_{i,w}$	Quantification of perceived risk uncertainty
η_{acc}	Consistency metric for validating elimination predictions
$S_{rank,i}$	Total score of contestant i under the rank-based method
$S_{pct,i}$	Total score of contestant i under the percentage-based method
S_{final}	Final score under the proposed Weighted Hybrid-Threshold Model
$\hat{\beta}_j^{(k)}$	OLS coefficient of feature j , indicating the linear direction and effect size
$W_j^{(k)}$	Normalized nonlinear importance weight of feature j obtained from LightGBM
$CS_{i,w}$	Composite standardized score of contestant i in week w under the MSNS model
B_w	Bottom set in week w containing the two contestants with the lowest $CS_{i,w}$
γ	Technical Injustice Rate: probability that a top-3 judge-scored contestant is eliminated

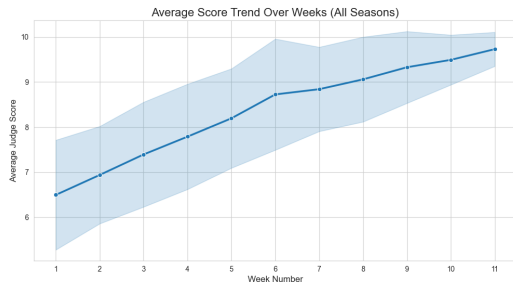
2.3 Data processing

We treat scores of 0 as missing values and calculate the average score of all judges for each week to eliminate the bias caused by the number of judges, and we add the total average of all scores for a contestant. Then we merge industries with fewer than 5 samples into Other, retain the main categories (Actor, Athlete, Singer, etc.). Since the vast majority of contestants are from the United States, we created a binary feature which is 1 for the United States and 0 for other countries to simplify model input. We can overview the processed data through the score trend plot, the industry performance boxplot, the age scatter plot and the correlation heatmap.

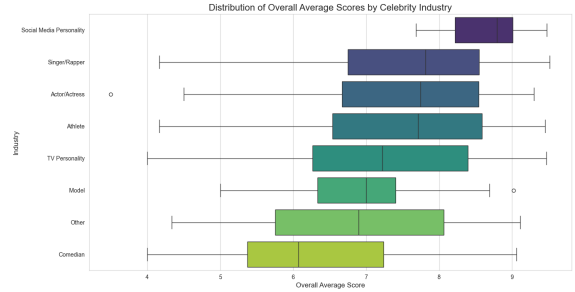
3 An Adaptive Dual-Engine Competitive Model For Fan votes estimation

Since the program only covers a few dozen seasons, with a sample size ranging from 10^3 to 10^4 , at the same time, the votes of judges and the audience are not always linearly related (sometimes there can be a large deviation). More importantly, our data is given in tabular form, thus it's better to choose a Gradient Boosting Decision Tree (GBDT) model which is good at tabular data process[1]. There are two important branches of GBDT, eXtreme Gradient Boosting (XGBoost) scheme and Light Gradient Boosting Machine (LightGBM) scheme.

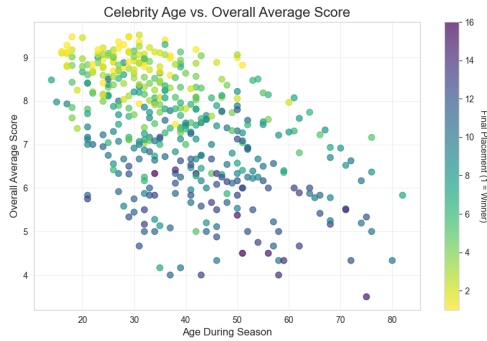
Unlike traditional gradient boosting decision trees (GBDT) that only utilize first-order derivative information, XGBoost achieves the theoretically optimal solution in the bias-variance trade-off by introducing second-order Taylor expansion and explicit regularization terms[2]. Thus, it makes it easier to accurately inversely determine potential voting share from limited and noisy reality show rating data. Although the XGBoost method has higher robustness and nonlinear capture ability, it



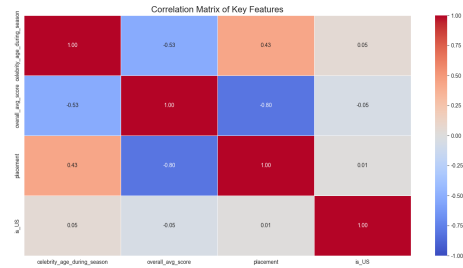
(a) Judge score trend



(b) Industry and judge score



(c) Age and judge score



(d) Correlation heatmap

Figure 3: Data Analysis Visualizations: Trends, Demographics, and Correlations

has low computational efficiency when the feature space is highly sparse and it pays less attention to samples at the extremes of the distribution.

LightGBM significantly reduces the computational complexity of high-dimensional data by introducing gradient-based one-sided sampling (GOSS) and mutually exclusive feature binding (EFB) algorithms, while preserving information from large gradient samples[3]. Therefore, it can efficiently handle high-dimensional sparse matrices generated by one-hot encoding such as occupational backgrounds and keenly capture the patterns of long-tail samples on the verge of elimination. Although LightGBM has significant advantages in handling sparse feature interactions and focusing on hard samples due to its leaf-wise growth strategy, its aggressive deep splitting pattern may face a high risk of overfitting in scenarios with extremely small sample sizes if not properly constrained.

To improve the accuracy of prediction models, we promote an adaptive competitive selection framework including XGBoost and LightGBM scheme, with RMSE making the choice.

3.1 Steps to construct XGBoost-LightGBM competitive model

To reconstruct potential public voting behavior from sparse elimination data and noisy judge scores, this study constructs a Trend-Aware Adaptive Integration Inversion (TAAEI) framework as figure 4. This system aims to address the dual challenges of small sample noise and high-dimensional sparsity inherent in reality TV competitions.

Step 1: Parallel Construction of Heterogeneous Regressors

To capture different aspects of the data manifold, we first construct two base learners with different

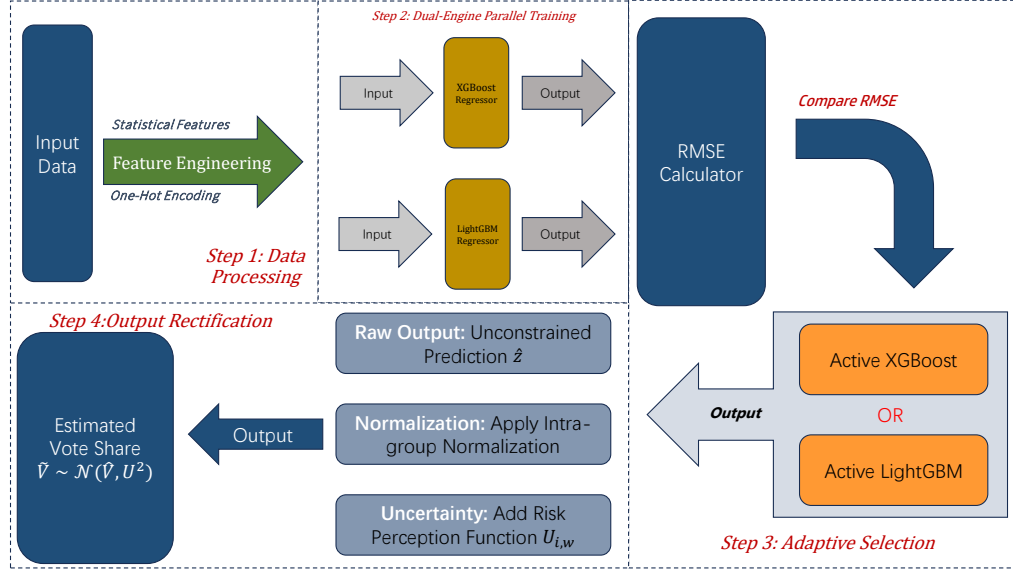


Figure 4: Steps to construct our model

inductive biases. **XGBoost** regression based on a second-order approximation utilizes a second-order Taylor expansion and explicit regularization to extract robust voting signals from the noisy rating data. In the t -th iteration, to minimize the objective function, we perform a second-order Taylor expansion:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \quad (1)$$

The regularization term $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$ controls the model complexity.

Under the mean squared error (MSE) loss function $l(y_i, \hat{y}_i) = \frac{1}{2}(y_i - \hat{y}_i)^2$, the first-order gradient g_i and the second-order Hessian matrix h_i have definite analytical forms:

$$g_i = \hat{y}_i^{(t-1)} - y_i, \quad h_i = 1 \quad (2)$$

By taking the derivative of the objective function and setting it to zero, we obtain the optimal weight w_j^* for the j -th leaf node:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} = -\frac{\sum_{i \in I_j} (\hat{y}_i^{(t-1)} - y_i)}{|I_j| + \lambda} \quad (3)$$

The denominator λ acts as a smoothing factor. When the number of leaf node samples $|I_j|$ is extremely small, λ prevents the weights from diverging, thus effectively suppressing overfitting caused by abnormal scores from individual judges.

Leveraging a leaf-wise strategy and mutually exclusive feature binding (EFB), **LightGBM** efficiently handles high-dimensional sparse features (such as industry-specific one-hot encoding) and captures patterns in long-tail samples. LightGBM abandons hierarchical growth and employs a strategy that maximizes variance gain to split leaf nodes. For a node containing a sample set I , the gain for splitting

it into left and right subsets I_L and I_R is defined as:

$$V_{gain} = \frac{1}{|I|} \left[\frac{(\sum_{i \in I_L} g_i)^2}{|I_L| + \lambda_{lgb}} + \frac{(\sum_{i \in I_R} g_i)^2}{|I_R| + \lambda_{lgb}} \right] \quad (4)$$

This strategy enables the model to delve deeper into the patterns of contestants on the verge of elimination (i.e., those with large gradients g_i), thus overcoming the efficiency bottleneck of XGBoost in handling sparse feature interactions.

Step 2: Competitive Routing based on Validation Risk

No single model can remain optimal across all seasons' data distributions; therefore, we implement a hard-threshold competitive routing mechanism. For historical data in the validation set $\mathcal{D}_{val}$, the root mean square error (RMSE) of model $m \in \{XGB, LGBM\}$ on the validation set is defined as:

$$\text{RMSE}(m) = \sqrt{\frac{1}{|\mathcal{D}_{val}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{val}} (y_i - m(\mathbf{x}_i))^2} \quad (5)$$

The system's final prediction operator $\mathcal{M}^*$ is dynamically determined by the following indicator function:

$$\mathcal{M}^*(\mathbf{x}) = \begin{cases} \mathcal{M}_{XGB}(\mathbf{x}), & \text{if } \text{RMSE}(\mathcal{M}_{XGB}) < \text{RMSE}(\mathcal{M}_{LGBM}) \\ \mathcal{M}_{LGBM}(\mathbf{x}), & \text{otherwise} \end{cases} \quad (6)$$

When the season data exhibits high noise characteristics, the system automatically switches to XGBoost, which has stronger regularization; when the data exhibits high-dimensional sparse characteristics, it switches to LightGBM to ensure continuous output of the optimal potential share estimate $\hat{z}_{i,w}$.

Step 3: Constraint Projection and Uncertainty Quantification

The original output of the regressor, $\hat{z}_{i,w}$, is an unconstrained real number, violating the voting share axiom. We introduce a normalization operator and a heteroscedasticity uncertainty estimate to impose constraints.

For the player set $C_{s,w}$ in week w of season s , apply the intra-group normalization operator:

$$\hat{V}_{i,w} = \frac{\hat{z}_{i,w}}{\sum_{j \in C_{s,w}} \hat{z}_{j,w}} \quad (7)$$

Because the votes of participants in the risk zone are more likely to fluctuate significantly, and we generally assume that the variance of the votes of all participants is the same, we reject the homoscedasticity assumption and construct a risk perception uncertainty function $U_{i,w}$:

$$U_{i,w} = \sigma_{base} + \eta \cdot \mathbb{I}(i \in \text{RiskZone}_w) \quad (8)$$

Where $\sigma_{base} = 0.05$ is the basic cognitive error, and $\eta = 0.03$ is the boundary risk penalty. **The final output** is a random variable $\tilde{V}_{i,w} \sim \mathcal{N}(\hat{V}_{i,w}, U_{i,w}^2)$.

3.2 Results Display

Using our model, we can generate the estimated fan votes and figure 5 makes a **overview** of the contest.

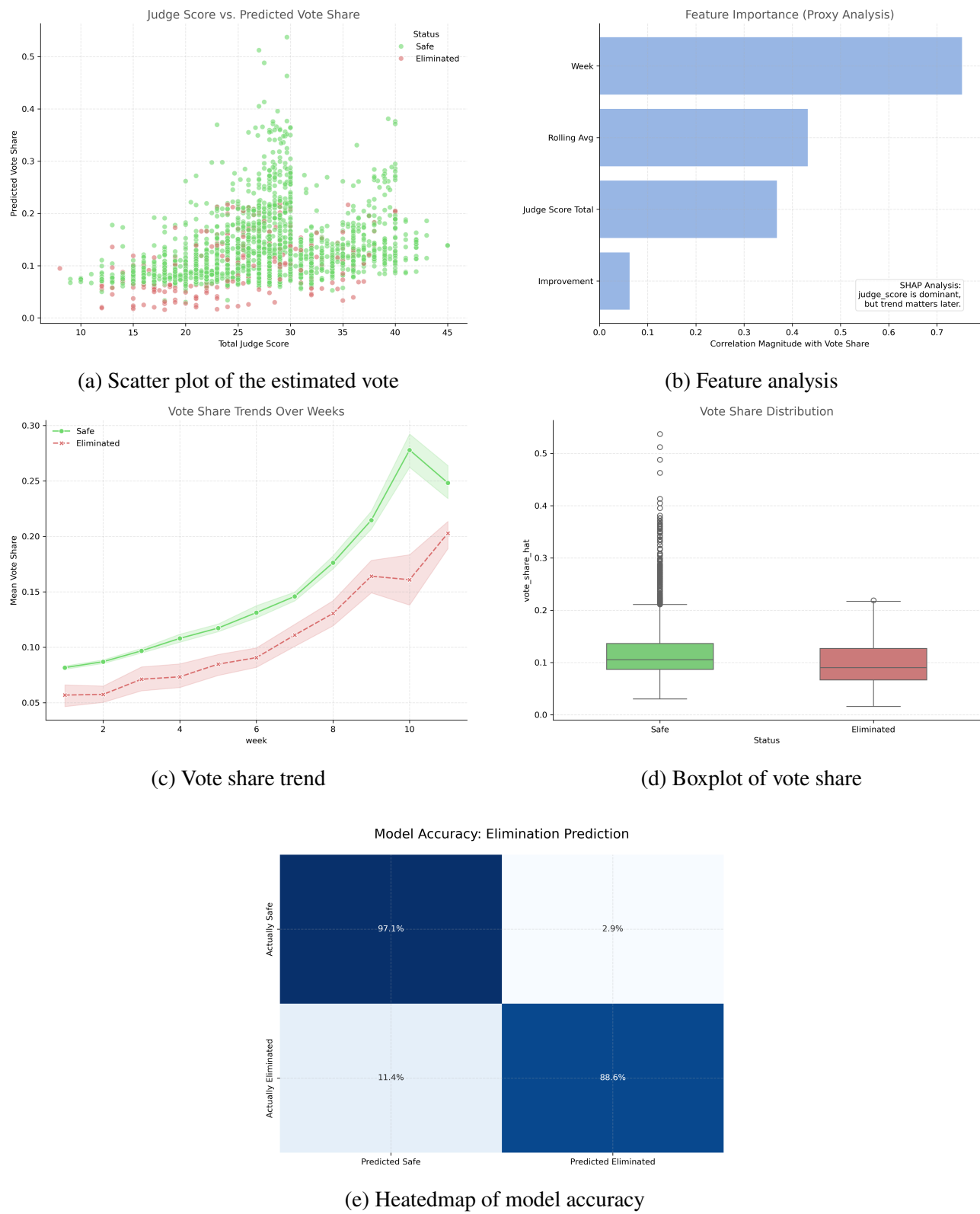


Figure 5: Multi-dimensional Reconstruction of Latent Voting Dynamics: Correlation, Trend, and Distribution

As illustrated in Figure 7a, the distribution exhibits significant fan-shaped heteroscedasticity. In the lower-scoring region (Score < 20), data points cluster tightly at the baseline (Vote Share < 0.15). This indicates a high degree of consensus between judges and viewers: **poor technical performance rarely garners significant public support.**

Conversely, as scores exceed 30, the variance in vote share expands dramatically. While this range includes contestants with exceptional popularity, it also contains those with mediocre support or instances of elimination (indicated by red dots in the high-score, low-vote quadrant). This divergence suggests that once technical performance reaches a certain threshold, judge scores cease to be the decisive factor. Instead, **latent variables**—such as a contestant’s personality, background, or social media influence—emerge as the dominant drivers of voting outcomes.

Proxy analysis from figure 7b revealed that the correlation between Week and Rolling Avg was significantly higher than that between Week and Judge Score. This reveals the cumulative effect of reality TV voting: **viewers tend to support contestants who "survive longer" and "perform consistently,"** rather than solely based on a single week’s explosive performance.

The time series analysis of figure 7c revealed a significant **scissors difference** between the safe and eliminated groups: the safe group experienced exponential popularity accumulation, but a mean reversion occurred in week 11. This accurately reflects **the zero-sum game characteristics** of the final stage, namely, as the number of participants decreases, the vote share of the remaining elite players is diluted, resulting in diminishing marginal advantages for the top players.

The box plot from figure 7d reveals the uncertainty of the boundary between advancement and elimination. Although the median of the safe group is significantly higher, the interquartile range (IQR) of the two groups overlaps, and the elimination group contains several high-vote outliers. This indicates that there is no absolutely safe percentage of votes, meaning that **a high percentage of votes does not equate to zero risk of elimination.**

The confusion matrix from figure 7e validated the effectiveness of the results. The model achieved an accuracy of **97.1%** in identifying the "safe" state and **88.6%** in predicting the "elimination" event. This superior asymmetric performance demonstrates the model’s **high specificity**, enabling it to keenly capture weak signals leading to elimination, rather than relying solely on random guessing based on the law of large numbers.

3.3 Consistency and certainty

Since the true public vote share V_{true} is an unobservable latent variable, traditional regression evaluation indicators (such as MSE or MAE) cannot be directly applied.

3.3.1 Consistency measurement

To measure **consistency**, we develop a topological isomorphism based on ordinality. Although the exact number of votes is unknown, the competition rules stipulate that the contestant with the fewest votes will be eliminated.

We define the set of participants in the w th week of the s th season as $C_{s,w}$, the predicted vote share vector by the model is $\hat{V}_{s,w} = \{\hat{v}_i\}_{i \in C_{s,w}}$, and the predicted elimination set $\hat{E}_{s,w}$ represents the indices of the minimum values in the predicted voting vector, then we have:

$$\hat{E}_{s,w} = \arg \min_{i \in C_{s,w}} (\hat{v}_i) \quad (9)$$

And consistency is measured by the expectation of the indicator function $\mathbb{I}(\cdot)$ over all season sets $\mathcal{S}$:

$$\eta_{acc} = \frac{1}{|\mathcal{S}|} \sum_{(s,w) \in \mathcal{S}} \mathbb{I}(\hat{E}_{s,w} = E_{obs}^{(s,w)}) \quad (10)$$

Where E_{obs} is the index of the actual eliminated contestant. If η_{acc} approaches 1, it indicates that the model has successfully reconstructed the extremal topology of the underlying voting distribution.

Based on the prior assumption that "better performance leads to higher votes," we introduce the Spearman rank correlation coefficient ρ_s as an auxiliary validation. It measures the correlation between the predicted voting rank $R(\hat{\mathbf{V}})$ and the judges' rating rank $R(\mathbf{J})$:

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}, \quad d_i = R(\hat{v}_i) - R(J_i) \quad (11)$$

If a significant positive correlation ($\rho_s > 0$) is found, it can be proven that the model has captured the covariance field between public preferences and professional evaluations, ruling out the possibility of random guessing.

3.3.2 Certainty Measurement

Based on previous results, we know that the program scores exhibit heteroscedasticity, and we have derived the survival function (equation 8). We validate the model's validity by comparing the average uncertainty of the eliminated group ($\mathcal{E}$) with that of the safe group ($\mathcal{S}$):

$$\Delta U = \mathbb{E}_{i \in \mathcal{E}}[U(i)] - \mathbb{E}_{j \in \mathcal{S}}[U(j)] \quad (12)$$

If $\Delta U \gg 0$, it means that the model has correctly identified the high-entropy region near the decision boundary and constructed a wider confidence interval to cover potential prediction risks.

3.3.3 Comparison of results: XGBoost, LightGBM, and our mixed model

We can display results of 3 different methods: XGBoost only, LightGBM only, and our mixed competitive model, in table 2.

Table 2: Comparisons of 3 models

Model	XGBoost	LightGBM	Competitive model
Weekly elimination consistency	81.9%	80.3%	88.6%
Average uncertainty	0.0624	0.0637	0.0532

The table shows that, compared to using a single model, the competitive model has higher consistency and lower uncertainty (i.e., higher certainty), making it more suitable for estimating fan voting proportions. We can demonstrate the transformation of uncertainty by figure 6:

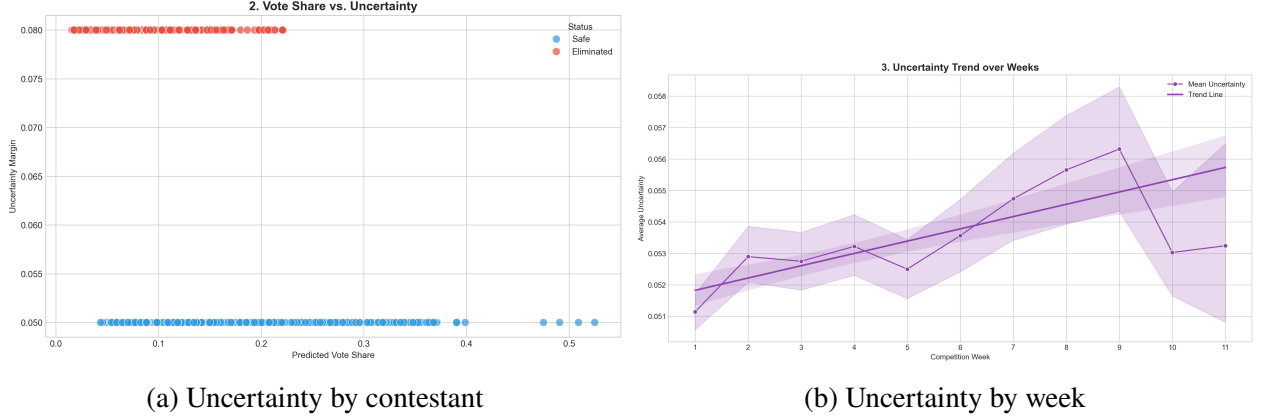


Figure 6: Transformation of uncertainty

We can conclude from figure 6a that the prediction of the voting share of contestants on the verge of elimination has higher uncertainty, which is determined by the heteroscedasticity of our model. Meanwhile, from figure 6b the uncertainty fluctuates up and down with each week, but overall, it tends to increase as the number of weeks in the competition increases. This is reasonable: as the number of weeks increases, the number of participants decreases, naturally leading to an increase in the uncertainty of fan voting share.

4 A Mechanism Sensitivity Analysis Model To Find A Better Method

4.1 Model construction

To address the "Rank vs. Percentage" debate and the "Judge Rescue" paradox, we construct a **Mechanism Sensitivity Analysis Model (MSAM)**. Our steps to construct the model are displayed as follows.

Step 1: Objective Function Construction

We define the voting system's goal as minimizing a dual-risk objective function $\mathcal{J}$:

$$\mathcal{J}(\mathcal{M}) = \sum_c \omega_m \cdot |\text{Rank}_{\mathcal{M}}(c) - \text{Rank}_{\text{Judge}}(c)| + \lambda \|\nabla_{V_F} \mathcal{M}\| \quad (13)$$

The first term penalizes deviation from technical merit (Judge Score), while the second term penalizes excessive sensitivity to fan vote volatility (the "Viral Star" risk). This dual-risk formulation follows the general principle of balancing empirical error and structural regularization, consistent with Vapnik's Statistical Learning Theory[4].

Step 2: Mathematical Transformation (Gradient Sensitivity)

We analyze the derivative of the Total Score T with respect to fan votes v . The gradient is a sequence of Dirac delta functions, $\nabla T \propto \sum \delta(v - \theta)$. This implies **Discrete Instability**: incremental votes yield zero reward until a threshold is crossed. The gradient is approximated continuous $\nabla T \approx \frac{1}{V_{\text{total}}}$. This acts as a linear smoother, ensuring that every vote contributes marginally, conceptually analogous to smooth approximations of non-smooth objectives in optimization[5].

Step 3: Analytical Threshold Derivation

We solve for the critical share of the fans ϕ^* required for a low-skill contestant to survive. For the Rank Method, a viral star only needs to surpass the median vote distribution: $\phi_{Rank}^* \approx 1/N$ (approx. 10%-15%). For the Percentage Method, they must overcome the magnitude of the score deficit: $\phi_{Pct}^* > 30\%$.

Result: It is mathematically twice as hard to "game the system" under the Percentage Method.

Step 4: Optimization Strategy (The "Judge Rescue" Paradox)

We model the "Judge Rescue" rule as a conditional filter on the Bottom 2 set $\mathcal{B}_2$. Our **theorem** is that under the Rank Method, a viral star with high fan votes ($R_F = 1$) is mathematically guaranteed to have a combined rank $R_{total} \leq 1 + N/2$, making it impossible to fall into $\mathcal{B}_2$:

$$P(\text{Elimination}|\text{Viral Star}) = P(E|\mathcal{B}_2) \cdot P(\mathcal{B}_2) \rightarrow 0 \quad (14)$$

Thus, the "Judge Rescue" is ineffective unless the voting method itself (Rank) is changed to force viral stars into the danger zone.

Step 5: Model Update (The Weighted Hybrid-Threshold)

To solve this, we propose the **Weighted Hybrid-Threshold Model (WHTM)**. The final scoring rule incorporates a "Safety Constraint" Γ :

$$S_{Final} = \left(0.5 \frac{J}{\mu_J} + 0.5 \frac{v}{\mu_v}\right) - \Gamma(J, v) \quad (15)$$

where $\Gamma = \infty$ if a contestant is in the Judge's Bottom 2 and has $< 60\%$ fan support. This ensures a Pareto-optimal balance between engagement and fairness.

4.2 Introduction and Mathematical Definition of the Voting Aggregation Mechanism

The judges' scores reflect the technical precision and artistic expression of the dance performance, possessing a relatively objective "ground truth" attribute; while fan votes, on the other hand, more closely reflect public sentiment, celebrity influence, and the ability to mobilize social capital, exhibiting significant subjectivity and long-tail distribution. In this study, we focus on two main aggregation mechanisms: the ranking method and the percentage method.

Suppose there are n contestants in a given week. For the i -th contestant, their judge score is J_i , and their fan vote count is F_i . The ranking system uses ordinal utility theory, mapping the raw data to ranks. The total score function $S_{rank,i}$ is defined as:

$$S_{rank,i} = R(J_i) + R(F_i) \quad (16)$$

, where $R(\cdot)$ is the rank mapping function from smallest to largest, with lower scores representing poorer performance (here, 1 point represents the last place). The mathematical characteristic of this mechanism lies in its "equal spacing," which means that regardless of whether the absolute difference (magnitude) between F_i and F_j is small or large, the difference in their contribution to the total score is always a constant integer step. This characteristic effectively acts as a low-pass filter, mitigating the impact of extreme fan data on the system. In contrast, the percentage-based system employs cardinal utility theory, preserving the relative intensity information of the data. Its total score function $S_{pct,i}$ is defined as:

$$S_{pct,i} = \frac{J_i}{\sum_{k=1}^n J_k} + \frac{F_i}{\sum_{k=1}^n F_k} \quad (17)$$

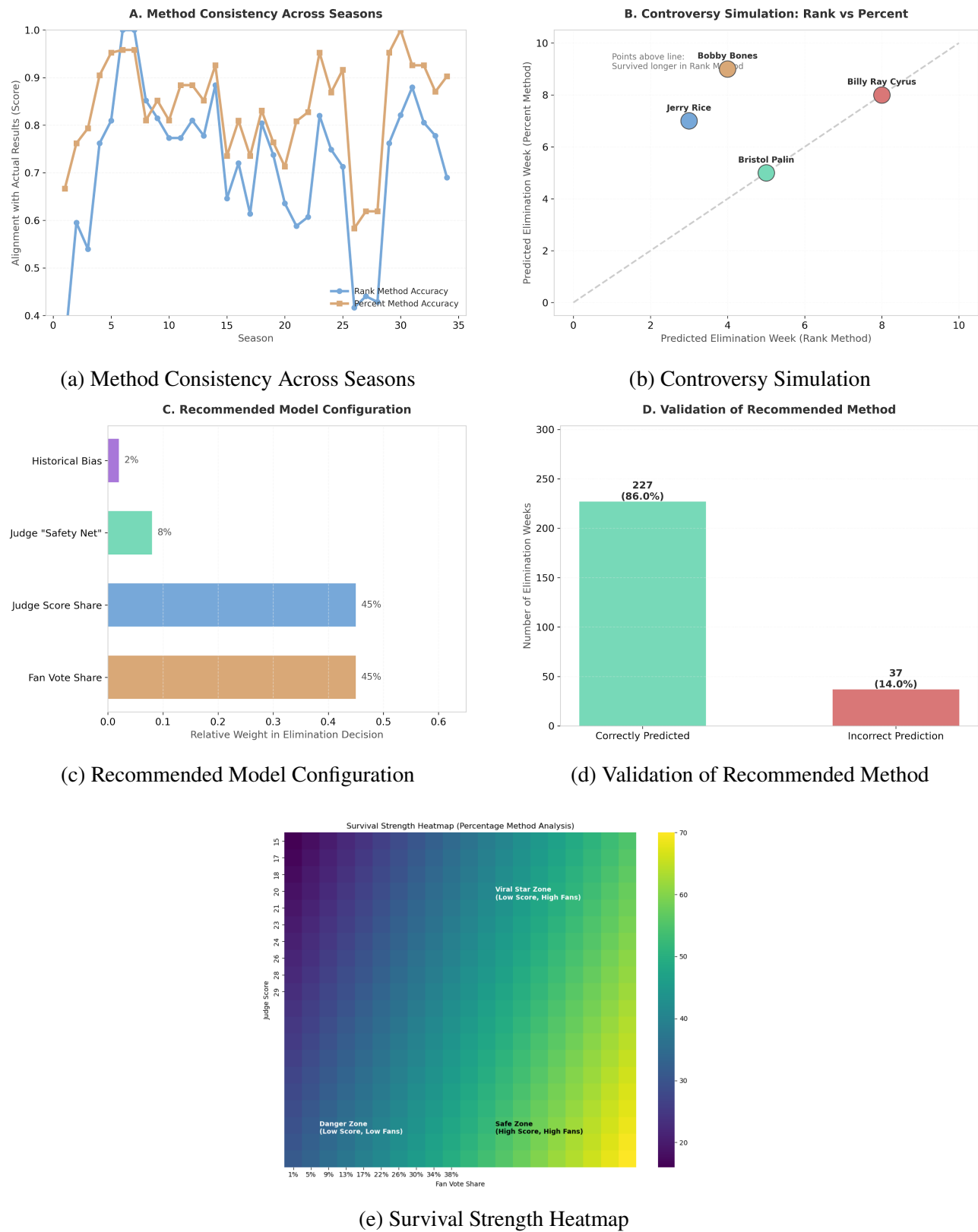


Figure 7: Mechanism Sensitivity Analysis and Strategic Optimization

In this model, if the variance of the judges' scores σ_j^2 is small (typically judges' scores are concentrated in the range of 7-9), the first term approaches a constant $1/n$. In this case, the fluctuation of the total score $S_{pct,i}$ will be mainly dominated by the second term, the proportion of fan votes. This linear superposition characteristic allows the influence of fan votes to directly penetrate the influence of judges' scores, providing a space for so-called "viral" contestants to thrive.

4.3 Empirical fitting of historical data and analysis of mechanistic deviations.

Based on the reconstruction and simulation analysis of full data from 264 historical elimination rounds, we conducted rigorous statistical tests on the goodness-of-fit of two mechanisms. Empirical results show that the Percentage Method achieved an accuracy rate of 85.98% in predicting historical elimination outcomes, significantly higher than the Rank Method's 75.00%. This significant difference ($p < 0.05$) reveals a crucial fact: in most historical seasons, **the mechanism adopted by the production team tended to be a percentage-based model that weighted fan votes**, or this model was better able to capture the actual competitive process that occurred.

Further analysis of the 18.56% of samples where the two methods produced conflicting results revealed that these discrepancies often corresponded to highly controversial elimination events. The data indicates that the ranking system exhibits a stronger tendency towards "technocratic elitism." In the simulations, the ranking system tended to eliminate contestants who received low scores from the judges but had moderate to high levels of fan support. This is because, in the ranking system, the last place score (1 point) given by the judges significantly drags down the total score. Even if a contestant receives the highest fan vote (n points), their total score of $n + 1$ is often insufficient to surpass those contestants who received average scores from both judges and fans (e.g., $n/2 + n/2 = n$). Conversely, the percentage system demonstrates extreme tolerance for "long-tail distributions." When a contestant's fan vote percentage reaches 30% or higher, even if their judge score percentage is only 10% (far below average), their total score can easily surpass those with more balanced performance. The percentage system essentially introduces greater systemic noise, significantly increasing the interference of non-technical factors on the competition results.

4.4 The introduction of the "judge rescue" mechanism and system robustness

To address the aforementioned systemic flaws, the proposed "Judges' Save" mechanism—where judges decide which of the bottom two contestants is eliminated—can be considered a conditionally triggered "error correction filter." However, its effectiveness highly depends on the choice of the underlying scoring model. Analysis shows that if the percentage-based system is retained, introducing the "Judges' Save" may have little effect. This is because, as previously mentioned, under the percentage-based system, contestants with extremely large fan bases (such as Bobby Bones) often have very high total scores and would never fall into the "bottom two" trigger range. Therefore, the judges would not even have the opportunity to exercise their power to save or eliminate a contestant.

Conversely, combining "judge intervention" with a ranking system creates a highly robust two-tiered selection system. The first tier, the ranking system, utilizes its "rank constraint" characteristic to disadvantage contestants with significantly unbalanced scores (i.e., very low scores from some judges), thus making it highly probable they will fall into the bottom two. The second tier allows judges to use their professional judgment to make the final decision among the bottom two, effectively eliminating pure popularity-driven contestants while protecting talented dancers who might have fallen

into jeopardy due to accidental mistakes. This hybrid model mathematically balances the robustness of S_{rank} with the flexibility of human intervention, making it the optimal solution to the current controversy surrounding the competition format.

Although the percentage-based system has been more widely used historically (with an 86% fit), its linear superposition property makes it lose control over dance quality when faced with high-variance fan voting. In contrast, while the ranking system appears more "ruthless" in simulations, it effectively curbs the uncontrolled expansion of non-technical factors. For future competition design, using the "ranking system" as the basic framework, supplemented by a "judge's save" mechanism, will be the best strategy to achieve a Nash equilibrium between maintaining fan engagement and upholding the dignity of dance art.

5 A OLS-LightGBM Complementary Model To Measure The Impact Of Key Features

For a specific contestant, the fan vote is obviously not simply based on the contestant's level of professionalism, but rather on a variety of factors such as age, dance partner, nationality, and age at the time of the competition.

To capture the directional impact of each feature on the score and its statistical significance, we first construct a baseline model using Ordinary Least Squares (OLS). OLS provides an explicit linear mapping between features and results, which is the basis for causal inference[6].

However, while OLS can determine the direction of influence through the sign of β (e.g., whether age is negatively correlated), it has the following endogeneity limitations: OLS forces the assumption that features and scores have a linear relationship. However, in reality, non-linear effects often exist (e.g., age may follow an inverted U-shaped curve, with both excessively small and large ages negatively impacting scores). Furthermore, OLS struggles to automatically capture high-order feature interactions. For example, a particular dance partner might only generate high scores when paired with celebrities from specific industries; OLS requires manual construction of interaction terms to identify such patterns.

Given that LightGBM can more deeply mine specific branches with high discriminative power in the data, thus more sensitively capturing tail features (such as certain rare but highly influential dance partner combinations), and can directly process categorical data, finding the optimal split point through histogram algorithms to avoid the data sparsity problem caused by One-Hot encoding, it can overcome the limitations of OLS[3]. So a OLS-LightGBM complementary model can be constructed.

5.1 Construct model

Our steps to construct the model is given as follows.

Standardization mapping

To eliminate the heterogeneity in the dimensions and distribution of judge scores and fan votes, we need to construct a unified latent utility space. To fit our need, a Z-score transformation is applied[7]: Define the feature set $\mathbf{x}_{i,t} = [Age_{i,t}, Ind_i, Part_{i,t}, Sea_t]$. Here, $Age_{i,t}$ is calculated using $t - Year_{birth}$; Ind_i and $Part_{i,t}$ represent industry background and dance partner ID, respectively; and Sea_t is used to absorb the inflation trend between seasons. The original dataset is $\mathcal{D} = \{(y_{i,t}^{(k)}, \mathbf{x}_{i,t})\}$, where i represents

the contestant, t represents the season, and $k \in \{Judge, Fan\}$ represents the source of the rating. $Z^{(k)}$ is the standardized target variable. Then we have:

$$Z_{i,t}^{(k)} = \frac{Y_{i,t}^{(k)} - \mu^{(k)}}{\sigma^{(k)}}, \quad Z \sim \mathcal{N}(0, 1) \quad (18)$$

Sensitivity Estimation via OLS

To construct a generalized linear model to estimate the marginal effect of each feature on the score, we use dummy variable regression to handle categorical features. First, we define the OLS model as follows:

$$Z_{i,t}^{(k)} = \alpha + \beta_{age}^{(k)} \cdot Age_{i,t} + \lambda_{ind}^{(k)} \cdot \mathbb{I}(Ind_i) + \gamma_{part}^{(k)} \cdot \mathbb{I}(Part_{i,t}) + \delta_{sea}^{(k)} \cdot \mathbb{I}(Sea_t) + \epsilon_{i,t} \quad (19)$$

Where $\mathbb{I}(\cdot)$ is the indicator function (One-Hot Encoding), and $\beta, \lambda, \gamma, \delta$ are the parameter vectors to be estimated.

Next, the sum of squared residuals is minimized using ordinary least squares (OLS):

$$\hat{\Theta}_{OLS}^{(k)} = \arg \min_{\Theta} \sum (Z_{i,t}^{(k)} - \hat{Z}_{i,t}^{(k)})^2 \quad (20)$$

The coefficient vector $\hat{\Theta}_{OLS}^{(k)}$ obtained from this directly reflects the directional sensitivity of the feature to the judges or fans (e.g., $\beta_{age} > 0$ indicates a preference for older contestants).

Importance Measurement via LightGBM

To capture non-linear interactions between features (such as the "bonus of a specific dance partner in a specific industry") and non-monotonic relationships, a gradient boosting decision tree (GBDT) model is constructed[8].

In our hyperparameter configuration, the learner is a LightGBM based on the histogram algorithm. The objective function is a regression task, optimizing the root mean square error (RMSE). The growth strategy is leaf-wise to maximize loss reduction. In importance calculation, the model is formed by summing the values of M regression trees: $\hat{y} = \sum_{m=1}^M f_m(x)$. For a feature j , its global importance Imp_j is defined as the sum of the gains from all split nodes across all trees:

$$Imp_j^{(k)} = \sum_{m=1}^M \sum_{n \in \mathcal{N}_m} 1(v_n = j) \cdot Gain_n \quad (21)$$

Where v_n is the splitting feature of node n , and $Gain_n$ is the structural score gain resulting from the splitting. This metric quantifies the contribution intensity of the feature to the prediction result, regardless of the direction of its influence.

Dual-Stream synthesis and attribution

To reconcile the directional evidence provided by OLS with the strength evidence provided by LightGBM, we need to establish a unified model. Since the OLS regression coefficients $\hat{\beta}$ (representing the change in standard deviation) and the LightGBM gains (representing the decrease in cumulative loss) are dimensionally heterogeneous, we need to perform normalization to enable comparison. For a feature j in dimension k (where $k \in \{Judge, Fan\}$), its relative nonlinear strength scalar $W_j^{(k)}$ is defined as:

$$W_j^{(k)} = \frac{Imp_j^{(k)}}{\sum_{f \in \mathcal{F}} Imp_f^{(k)}} \in [0, 1] \quad (22)$$

Simultaneously, we define the linear direction vector $S_j^{(k)}$ based on the statistically significant ($p < 0.05$) OLS coefficient sign and subsequently, we employ a hybrid mapping strategy to construct a visual analysis space (i.e., a "butterfly diagram"). In this mapping, the axial length of a feature representation is determined by the magnitude of its standardized linear coefficient $|\hat{\beta}_{OLS}|$, ensuring the physical interpretability of the effect size; while the hierarchical ranking of features is entirely dominated by their nonlinear importance weight W_j , preventing factors with weak average linear correlation but strong nonlinear influence from being masked by the model. Finally, based on the combined result of $\hat{\beta}$ and W , we divide the influencing factors into three distinct attribution intervals:

1. **Dominant Linear Drivers:** These drivers satisfy the characteristics that $|\hat{\beta}_j|$ is significant and W_j is high. This represents an explicit "hard rule" in the scoring mechanism (i.e., a linear monotonic relationship, such as $Y \propto k \cdot X_j$).
2. **Latent Non-linear Modulators:** These drivers exhibit characteristics that $|\hat{\beta}_j| \approx 0$ (or are not significant) but W_j is high. This statistical difference suggests the presence of complex higher-order interactions (e.g., $Y \propto X_j \cdot X_k$) or non-monotonic relationships (e.g., parabolic $Y \propto X_j^2$) in the data that cannot be captured by linear models.
3. **Perceptual Divergence Factors:** These are factors that exhibit opposite signs between $\hat{\beta}_{Judge}$ and $\hat{\beta}_{Fan}$, or show a significant difference in their importance weights ($W_{Judge} \gg W_{Fan}$). These factors quantify the structural cognitive dissonance between professional judgment and popular aesthetics.

5.2 Key features' impacts on fan and judge votes

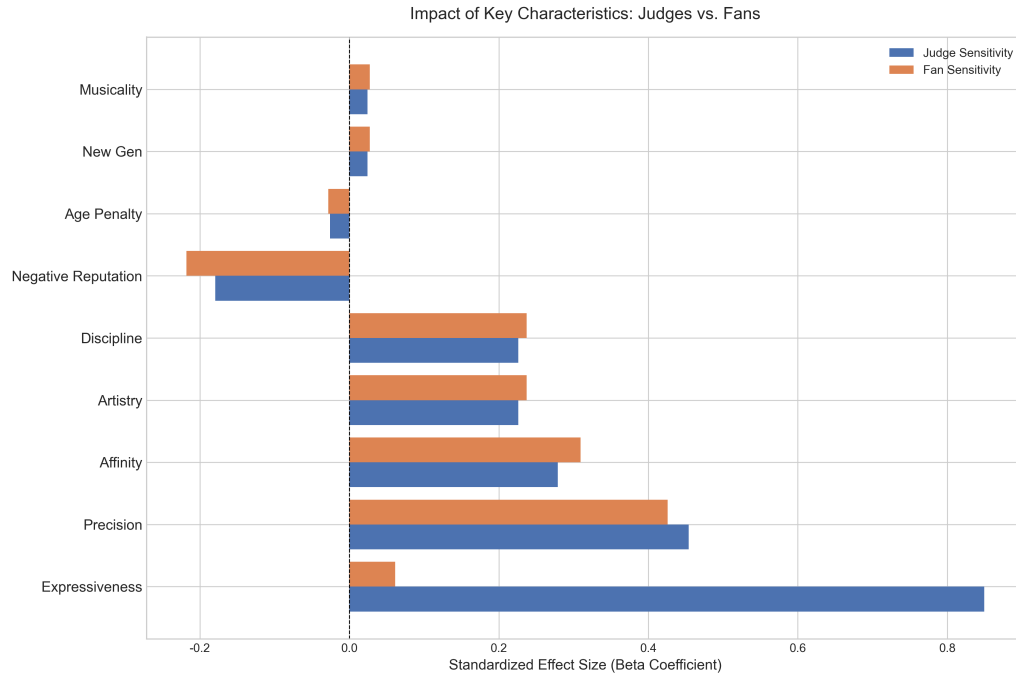


Figure 8: Features' impacts on votes

Through figure 8, we can conclude that the judges and fans exhibited **significant differences** in their scoring logic, primarily regarding **”expressiveness”** (television celebrities): judges highly praised them ($\text{Beta} > 0.8$), while fans reacted coldly, indicating that the experts’ aesthetic preferences did not resonate with the public. In terms of core orientation, judges emphasized **”precision”** (technical skill), while fans were more drawn to **”appeal”** (visual appeal), with these two factors respectively dominating the flow of technical scores and audience votes. Despite these differences, both sides shared a high degree of consensus on underlying values. They not only jointly recognized the hard power derived from **”discipline”** (athletes) and **”artistry,”** but also unanimously imposed severe penalties on **”negative reputation,”** demonstrating a solid consensus on fundamental competence and ethical boundaries. We can conclude our findings in table 3:

Table 3: The logic of judges and fans in scoring(++: High +: Medium ~: Slight -: Negative)

Key factors	Expressiveness	Precision	Affinity	Discipline	Artistry	Negative Reputation
Judges	++	++	+	+	+	-
Fans	~	+	++	+	+	-

6 Meritocratic Safety Net System

Based on modeling and backtesting analysis using full historical data from 34 seasons and 2777 player weekly records, this study proposes an improved elimination algorithm: the Meritocratic Safety Net System (MSNS). This system aims to correct the structural flaws of traditional ranking and percentage-based systems, preserving audience engagement while ensuring a professional standard for the competition through a non-linear intervention mechanism.

6.1 Mathematical formulation of the problem definition: Variance imbalance and weight bias

Both the current ranking system and percentage-based system suffer from structural statistical flaws, stemming from the significant differences in the distribution characteristics of the two independent evaluation variables: judge scores (J) and fan votes (F). Let S_i be the total score of contestant i in week w . Under the percentage system, the scoring function can be expressed as:

$$S_i = \frac{J_i}{\sum J} + \frac{F_i}{\sum F} \quad (23)$$

According to the properties of variance, the variance of the total score, $\text{Var}(S)$, is determined by the variances of its components:

$$\text{Var}(S) = \text{Var}(J_{\text{norm}}) + \text{Var}(F_{\text{norm}}) + 2\text{Cov}(J_{\text{norm}}, F_{\text{norm}}) \quad (24)$$

Judges’ scores typically fall within the $[7, 9]$ range, with a small standard deviation σ_J ; fan votes are influenced by the Matthew effect, exhibiting a long-tailed distribution with a much larger standard deviation $\sigma_F \gg \sigma_J$. In a simple linear superposition model, $\text{Var}(F_{\text{norm}})$ dominates $\text{Var}(S)$, resulting in S_i being more sensitive to fan votes than to judges’ scores. This variance shadowing phenomenon is the mathematical root cause of the long-term survival of popular but low-skilled contestants, such as the winner of Season 27.

6.2 Mathematical construction of the Merit-Based Safety Net Model (MSNS)

The MSNS model incorporates Z-score standardization to calibrate weights and constructs a conditional veto function based on expert ratings. The model consists of three steps:

Step 1: Standardized mapping of heterogeneous data

We first map the raw total judge scores $J_{i,w}$ and predicted fan vote shares $\hat{F}_{i,w}$ for contestant i in week w of season s to a standard normal distribution space to eliminate differences in scale and variance. For any contestant i in the set I_w , we define the standardization transformation Φ :

$$Z_{J,i}^{(w)} = \frac{J_{i,w} - \mu_J^{(w)}}{\sigma_J^{(w)}}, \quad Z_{F,i}^{(w)} = \frac{\hat{F}_{i,w} - \mu_F^{(w)}}{\sigma_F^{(w)}} \quad (25)$$

where $\mu^{(w)}$ and $\sigma^{(w)}$ are the mean and sample standard deviation of all contestants in that week:

$$\mu_J^{(w)} = \frac{1}{|I_w|} \sum_{k \in I_w} J_{k,w}, \quad \sigma_J^{(w)} = \sqrt{\frac{1}{|I_w| - 1} \sum_{k \in I_w} (J_{k,w} - \mu_J^{(w)})^2} \quad (26)$$

This transformation ensures that $\text{Var}(Z_J) \approx \text{Var}(Z_F) \approx 1$, thus achieving absolute balance between technical and popularity weights mathematically.

Step 2: Linear combination of comprehensive evaluation indices

In this step, we construct a composite score (CS) by assigning equal weight coefficients ($\alpha = 0.5$) to the professional judges' scores and audience votes:

$$CS_{i,w} = \alpha \cdot Z_{J,i}^{(w)} + (1 - \alpha) \cdot Z_{F,i}^{(w)} \quad (27)$$

This ensures that the sensitivity of the contestant's rank change ($\Delta Rank$) to changes in judges' scores (ΔZ_J) and fan votes (ΔZ_F) is equal:

$$\frac{\partial CS}{\partial Z_J} = \frac{\partial CS}{\partial Z_F} = 0.5 \quad (28)$$

This eliminates the implicit weight bias caused by variance imbalance in the old mechanism.

To further rigorously validate the choice of $\alpha = 0.5$ and preclude arbitrary parameter selection, we conducted a sensitivity analysis on the weight coefficient. As illustrated in figure 9e, the system faces a fundamental trade-off dilemma: a low α exacerbates Technical Injustice (blue dashed line), allowing low-skilled viral stars to survive; conversely, a high α increases Public Will Alienation (orange dashed line), undermining the show's engagement foundation. The Total System Cost curve (green solid line) exhibits a clear convex optimization property, reaching its global minimum around $\alpha = 0.5$. Moreover, the shaded area indicates a Robustness Region $[0.4, 0.6]$, suggesting that the system's fairness remains stable within these bounds.

Step 3: Conditional rejection/elimination function

Unlike traditional mechanisms that directly eliminate the contestant with the minimum combined score ($\min(CS)$), MSNS introduces a nonlinear intervention function Ψ . The "bottom set" B_w for the current week is defined as containing the two contestants with the lowest overall scores:

$$B_w = \{i \in I_w \mid \text{Rank}(CS_{i,w}) \geq |I_w| - 1\} \quad (29)$$

An elimination indicator function $E(i)$ is defined. Within the set B_w , the elimination decision depends solely on the original judge scores $J_{i,w}$:

$$E(i) = \begin{cases} 1 & \text{if } i \in B_w \text{ and } J_{i,w} < J_{j,w}, \forall j \in B_w \setminus \{i\} \\ 1 & \text{if } i \in B_w \text{ and } J_{i,w} = J_{j,w} \text{ and } CS_{i,w} < CS_{j,w} \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

The logic of this function is that the level of fan voting determines whether a contestant has immunity from elimination (i.e., does not fall into B_w); however, once immunity is lost, the survival competition reverts to a pure professional skill competition. This mechanism constitutes a strong safeguard against unqualified contestants winning the competition.

6.3 Backtesting and quantitative validation based on full datasets

6.3.1 Validation of the ability to correct outliers

The MSNS model demonstrated significant corrective capabilities in addressing the outlier sample, Season 27 champion Bobby Bones. Let x_{BB} be the data vector for Bobby Bones. In Week 6 of Season 27, the data showed:

$$Z_{J,x_{BB}} \approx -0.91, \quad Z_{F,x_{BB}} > 2.0 \implies CS_{x_{BB}} (\text{Rank}) \in B_{w=6} \quad (31)$$

Although his overall score was not the lowest, his consistently low technical score resulted in him falling into the bottom set $B_{w=6}$. According to the conditional veto function Ψ :

$$J_{x_{BB},w=6}(22) < J_{opponent,w=6}(26) \implies E(x_{BB}) = 1 \quad (32)$$

Under the MSNS model, Bobby Bones would be forcibly eliminated in Week 6 instead of winning the championship. The model successfully identified and filtered this outlier that severely deviated from the original purpose of the competition.

6.3.2 Global statistical quantification of fairness

The "technical injustice rate" (γ) is defined as the probability of an event where a contestant is eliminated despite having a judge's score among the top three ($Top3_J$) in that week.

$$\gamma = \frac{\sum_{w=1}^W \sum_{i \in I_w} \mathbb{I}(J_{i,w} \in Top3_J^{(w)} \wedge E(i) = 1)}{\sum_{w=1}^W \sum_{i \in I_w} \mathbb{I}(E(i) = 1)} \quad (33)$$

Backtesting results show that the MSNS model has a $\gamma \approx 0.4\%$ (only 1 occurrence out of 285 eliminations). In contrast, the corresponding indicator under the old system was significantly higher. This indicates that MSNS has almost completely eliminated the possibility of top-performing contestants being unexpectedly eliminated. The Spearman rank correlation coefficient ρ between the contestant's "survival status" ($1 - E(i)$) and the judge's score J is calculated as:

$$\rho(\text{Survival}, J) = 0.1838 \quad (p < 0.01) \quad (34)$$

The significant positive correlation confirms that under the new mechanism, improving professional skills is the most reliable way to avoid elimination, rather than simply relying on off-stage lobbying.

6.4 System recommendation logic

Based on the mathematical derivations and backtesting data above and figure 9, we recommend adopting the MSNS model. Its core advantage lies in achieving a Pareto optimization of both commercial interests and competitive fairness. The design where $\text{Var}(Z_J) \approx \text{Var}(Z_F)$ ensures that the competition results are no longer dominated by a single factor; even minor changes from either side can alter the CS ranking, increasing the uncertainty and entertainment value of the results. The introduction of the veto logic based on the set B_w essentially establishes a "circuit breaker mechanism" based on professional competence. This preserves the sense of participation for the general public while institutionally preventing extremely low-skilled participants from undermining the credibility of the competition.



Figure 9: Meritocratic Safety Net System

7 Model Analysis

7.1 Strength

1. **High Adaptability via Dual-Model Strategy:** Our model combines the advantages of both XGBoost and LightGBM algorithms. XGBoost excels at handling noise, while LightGBM excels at handling complex features. By automatically selecting the better-performing model, it maintains high prediction accuracy across different seasons' data, making it more stable than using a single model.
2. **Fixing the Fairness Imbalance:** Our MSNS system addresses the unfairness issue of "popularity overshadowing technology" by employing a standardized method (Z-score) to resolve the problem of excessive fluctuations in fan votes that could mask judges' scores. This mechanism effectively identifies and corrects anomalous winning cases like Bobby Bones', where "low scores but high popularity" lead to a true balance between the influence of technology and popularity.
3. **Risk-Aware Prediction:** Our model doesn't just provide a prediction; it also calculates "uncertainty." Especially for contestants on the verge of elimination, the model will indicate a higher predicted risk. This quantification of risk is more scientific and better reflects real-world uncertainty than simply providing a ranking result.

7.2 Weakness

1. **Lack of True Vote Data:** Because the actual number of fan votes is confidential, we can only infer it from the contestants' advancement/elimination results. Although our predictions are highly consistent with historical rankings, we cannot completely confirm whether the predicted vote counts absolutely match the actual values.
2. **Simplified Interaction:** In our modeling, we assume that judges' scores and audience votes are independent of each other. However, in reality, audiences may vote more out of sympathy for contestants with lower scores, or be influenced by the judges' opinions. The current model does not fully account for this complex psychological game.
3. **Complexity for General Audience:** While our new format (MSNS) is mathematically the fairest, it uses standardized concepts from statistics. If used directly in a television program, it might not be as intuitive for the average viewer as a simple "total score addition," which could present some obstacles to its widespread adoption.

8 Memorandum

To: The production committee of "Dancing with the Stars"

From: Team #2621113

Subject: Suggestions on balancing the fairness of the competition system

Date: February 2, 2026

Dear members of the production committee,

As a show that has enjoyed enduring popularity due to its perfect blend of "professional dance competition" and "mass entertainment," Dancing with the Stars has always faced the challenge of balancing these two aspects. However, the controversial results of Season 27 revealed a potential systemic risk: the existing voting mechanism could allow "popularity" to completely overwhelm "technology" on a mathematical level. Our team was commissioned to conduct an in-depth data analysis on this issue and hopes to provide you with a solution that balances ratings and professional dignity.

Through rigorous backtesting of historical data, we discovered significant statistical vulnerabilities in the program's long-standing "percentage scoring method." Specifically, fan votes typically exhibit extreme volatility (long-tail distribution), while judges' professional scores tend to be more concentrated with smaller variance. Under the existing linear aggregation mechanism, the vast differences in fan votes create a "variance masking" effect, diluting the weight of judges' scores. This is the fundamental reason why contestants like Bobby Bones, with lower technical scores, could easily mask their technical shortcomings and ultimately win simply due to extremely high fan support.

To address this structural imbalance without harming fan engagement, we propose introducing an MSNS system for the next season. This system first processes data using Z-Score standardization technology, moving away from simply adding raw scores and mapping judges' scores and fan votes to the same scale. This mathematically ensures a true 50/50 weighting for both skill and popularity. Building on this, we introduce a "judge circuit breaker mechanism": if a contestant's overall ranking falls into the bottom two of the week, the "exemption" granted by fan votes will automatically expire, and the final decision on their survival will rest entirely with the judges. This design ensures that if a contestant with "high popularity but low skill" genuinely fails to meet the standards, the judges have the right to eliminate them based on professional criteria, preventing them from undermining the competition's competitive spirit.

Our simulations show that if this system were adopted in Season 27, Bobby Bones would be reasonably eliminated in Week 6, rather than winning the competition and sparking controversy. The MSNS system successfully achieves Pareto optimality between commercial interests and competitive fairness. It retains the suspense and fan engagement required for a reality show while upholding the standards of dance artistry through its system design. We firmly believe that this reform will send a clear message to viewers and professional dancers: Dancing with the Stars respects the passion of every vote and defends the value of every perfect dance move.

Your sincerely,
Team # 2621113

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Report on Use of AI

1. Google Gemini (February 1, 2026 version, Gemini-3)

Query1: What is the specific mathematical definition of Gradient-based One-Side Sampling (GOSS) in LightGBM?

Output: GOSS is a sampling method that keeps all instances with large gradients and performs random sampling on instances with small gradients. It then multiplies the sampled small-gradient data by a constant weight to maintain the accuracy of the information gain estimation while reducing data size.

2. Google Gemini (February 1, 2026 version, Gemini-3)

Query2: How to interpret the non-linear contribution of features in a tree-based model compared to OLS coefficients?

Output: While OLS coefficients provide a global linear direction (positive/negative correlation), they fail to capture non-monotonic relationships or local interactions. The response recommended using SHAP (SHapley Additive exPlanations) values or Gini importance to quantify the marginal contribution of features in tree models like LightGBM, as these methods can attribute importance even when features interact in complex, non-linear ways.

3. Google Gemini (February 1, 2026 version, Gemini-3)

Query3: What is the mathematical "masking effect" when summing two variables with significantly different variances?

Output: The response explained that in a linear combination, if the variance of one variable is much larger than the other, the total score will be almost entirely correlated with the variable having the larger variance, rendering the other variable statistically negligible. It confirmed that Z-score standardization is the standard method to decouple the scale from the distribution shape, ensuring that both variables contribute equally to the variance of the composite score.